

# Model

## 1 Model

### 1.1 Environment

Consider a metropolitan area consisting of a finite set of locations  $S = \{1, 2, \dots, n\}$ . These locations are partitioned into municipalities  $g \in G$ :

$$S = \bigcup_{g \in G} S^g, \quad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location  $i \in S$  is endowed with a fixed supply of land  $\bar{H}_i$ , allocated between residential and commercial use by local zoning:

$$\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where  $\bar{H}_i^R$  is residential floor space capacity and  $\bar{H}_i^F$  is commercial floor space capacity. Both are chosen by the municipal government and assumed to be fully utilized in equilibrium. Land at each location is owned by absentee landlords who collect  $q_i^R \bar{H}_i^R + q_i^F \bar{H}_i^F$  and consume it outside the metropolitan area; this rental income has no further equilibrium feedback — it does not enter household budgets or municipal welfare.

#### 1.1.1 Transportation

Locations are connected by a transportation network  $(S, E)$ , where  $E \subseteq S \times S$  is a set of directed edges (links). Transportation investment  $I_{k\ell}$  is chosen at the *edge* level, for each  $(k, \ell) \in E$ . The utility cost of traveling through edge  $(k, \ell)$  is

$$d_{k\ell} = \exp(\kappa \times \text{time}_{k\ell}),$$

where  $\text{time}_{k\ell}$  is the travel time on edge  $(k, \ell)$  and  $\kappa > 0$  is a scale parameter, following Ahlfeldt et al., 2015. The travel time  $\text{time}_{k\ell}$  is a function of the edge-level traffic  $n_{k\ell}$  and investment  $I_{k\ell}$ :

$$\text{time}_{k\ell} = \delta_{kl}^0 + \delta_{kl}^1 \frac{n_{k\ell}^\rho}{I_{k\ell}^\beta}.$$

Following Fajgelbaum and Schaal, 2020,  $\delta_{kl}^0$  is the free-flow travel time on edge  $(k, \ell)$ ,  $\delta_{kl}^1 > 0$  is a scale parameter, and  $\rho, \beta > 0$  are elasticities of travel time with respect to traffic and investment respectively. Therefore for a given route  $r$ , the commuting cost  $\tau_{ij}$  between residence  $i$  and workplace  $j$  is the product of the costs of all edges along that route:

$$\tau_{ijr} = \prod_{(k, \ell) \in r_{ij}} d_{k\ell}(I_{k\ell}, n_{k\ell}),$$

where  $r_{ij}$  is the route chosen by commuters from  $i$  to  $j$ .

There are three types of agents: households, competitive firms, and governments. Households choose residence and workplace locations. Firms produce a freely tradable good using labor and commercial floor space. Government decisions operate at two levels: municipal governments choose local zoning, and a metropolitan transportation authority chooses investment across all links. In the decentralized equilibrium, municipalities and the transportation authority choose policies simultaneously, each treating the other's decisions as given.

## 1.2 Households

A household chooses a residence location  $i$ , a workplace  $j$ , and a route  $r$  from a finite, exogenously given choice set  $R_{ij}$  of feasible routes between  $i$  and  $j$  (e.g. the  $k$  shortest GIS paths, or distinct road/transit alternatives) to maximize utility. Preferences over the numaire good  $Q_{ij}$  and residential floor space  $h_{ij}^R$  are Cobb-Douglas:

$$u_{ijr}(\omega) = \frac{B_i(N_i, L_i)}{\tau_{ijr}} \left( \frac{Q_{ij}}{\alpha} \right)^\alpha \left( \frac{h_{ij}^R}{1-\alpha} \right)^{1-\alpha} v_{ij,r}(\omega),$$

where  $B_i(N_i, L_i) = \bar{B}_i / \phi(N_i, L_i) > 0$  is the effective residential amenity at location  $i$ : an exogenous location-specific level  $\bar{B}_i$ , discounted by the congestion disutility  $\phi(N_i, L_i)$  from local residential and workplace density (defined below in §Municipal Zoning).  $\alpha \in (0, 1)$  is the expenditure share on the composite good,  $\tau_{ijr}$  is the commuting cost along route  $r$ , and  $\{v_{ij,r}(\omega)\}$  is a vector of idiosyncratic shocks, jointly distributed according to a nested (generalized extreme value) Fréchet copula with cumulative distribution function

$$F(\{v_{ij,r}\}) = \exp \left[ - \sum_{ij \in S^2} T_i E_j \left( \sum_{r \in R_{ij}} v_{ij,r}^{-\theta} \right)^{\epsilon/\theta} \right], \quad \theta \geq \epsilon > 1. \quad (1)$$

The outer exponent is  $+\epsilon/\theta$ , not  $-\epsilon/\theta$ : a valid CDF must satisfy  $F \rightarrow 1$  as any  $v_{ij,r} \rightarrow \infty$  holding the rest fixed, which pins the sign down.  $T_i, E_j$  enter at the outer  $(i, j)$  nest only, not inside the inner route sum: they are location-pair accessibility scales, not route-specific ones, exactly as in the single-route model. Setting  $R_{ij} = \{r\}$  for every  $(i, j)$  collapses (1) to the original univariate Fréchet marginal  $\Pr(v_{ij} \leq z) = \exp(-T_i E_j z^{-\epsilon})$ , confirming the nesting strictly generalizes it;  $\theta \geq \epsilon$  is required for (1) to be a valid (positively correlated within-nest) GEV copula.

The household's budget constraint is

$$Q_{ij} + q_i^R h_{ij}^R \leq w_j,$$

where  $q_i^R$  is the residential rent at location  $i$  and  $w_j$  is the wage at workplace  $j$  — local congestion no longer enters the budget directly; it operates entirely through the amenity  $B_i(N_i, L_i)$  above. Optimal demands are  $Q_{ij}^* = \alpha w_j$  and  $h_{ij}^{R*} = (1-\alpha)w_j/q_i^R$ , giving the deterministic component of indirect utility for a given route,

$$\tilde{V}_{ijr} = \frac{B_i(N_i, L_i) w_j}{\tau_{ijr} (q_i^R)^{1-\alpha}}.$$

Residential floor space clears when aggregate demand from all residents of  $i$  exhausts the fixed supply  $\bar{H}_i^R$ : summing  $h_{ij}^{R*}$  over workplaces  $j$ , weighted by the population  $N\pi_{ij}$  commuting from  $i$  to  $j$ ,

$$\sum_j N\pi_{ij} \frac{(1-\alpha)w_j}{q_i^R} = \bar{H}_i^R \iff q_i^R = \frac{(1-\alpha) \sum_j N\pi_{ij} w_j}{\bar{H}_i^R}, \quad (2)$$

which pins down  $q_i^R$  as a function of the commuting-flow pattern  $\{\pi_{ij}\}$ , wages, and the zoning-determined residential supply  $\bar{H}_i^R$ .

**Upper nest choice** : applying the GEV choice-probability formula to (1), households compare residence-workplace pairs using the route-choice-corrected value  $\tilde{V}_{ij}$ , with  $T_i, E_j$  entering exactly as in the single-route model:

$$\pi_{ij} = \frac{T_i E_j \tilde{V}_{ij}^\epsilon}{\sum_{i'j' \in S^2} T_{i'} E_{j'} \tilde{V}_{i'j'}^\epsilon} = \frac{T_i E_j \tilde{V}_{ij}^\epsilon}{\Phi}, \quad (3)$$

where  $\tilde{V}_{ij} = \frac{B_i(N_i, L_i) w_j}{\tau_{ij}(q_i^R)^{1-\alpha}}$  and  $\tau_{ij} = \left( \sum_{r \in R_{ij}} \tau_{ijr}^{-\theta} \right)^{-1/\theta}$  is the Fréchet mean of route costs such that the population choosing  $i, j$  pair is:

$$N_{ij} = T_i E_j \tilde{V}_{ij}^\epsilon \frac{N}{\Phi}, \quad (4)$$

**Lower nest choice** : conditional on choosing  $(i, j)$ , the household chooses a route  $r \in R_{ij}$ . Following Allen and Arkolakis (2022), we can rewrite  $\tau_{ij} = \left( \sum_{r \in R_{ij}} \tau_{ijr}^{-\theta} \right)^{-1/\theta}$  such that

$$\tau = ((I - A)^{-1})^{-1/\theta}, \quad (5)$$

where  $A$  is a matrix with elements  $(k, l)$  is  $d_{kl}^{-\theta}$ , such that for matrix  $\tau$ , the element  $(i, j)$  is  $\tau_{ij}$ . Accordingly, we can derive the link intensity as follows:

$$\pi_{jk}^{kl} = \left( \frac{\tau_{ij}}{\tau_{ik} d_{kl} \tau_{lj}} \right)^\theta \quad (6)$$

Therefore I can derive the traffic on edge  $(k, l)$  as follows:

$$n_{kl} = \sum_{ij \in S^2} N_{ij} \pi_{jk}^{kl} \quad (7)$$

### 1.3 Firms

At each location  $j$ , competitive firms produce a freely tradable good using labor  $L_j$  and commercial floor space  $H_j^F$ :

$$Y_j = A_j L_j^\gamma (H_j^F)^{1-\gamma}, \quad (8)$$

where  $A_j > 0$  is location-specific productivity and  $\gamma \in (0, 1)$ . Taking wages  $w_j$  and commercial rents  $q_j^F$  as given, profit maximization yields:

$$L_j = \left( \frac{\gamma A_j}{w_j} \right)^{1/(1-\gamma)} H_j^F, \quad q_j^F = (1-\gamma) \left( \frac{\gamma}{w_j} \right)^{\gamma/(1-\gamma)} A_j^{1/(1-\gamma)}.$$

**Productivity agglomeration.** Local productivity is itself subject to a Marshallian externality from local employment density:

$$A_j = \bar{A}_j L_j^\zeta, \quad \bar{A}_j > 0, \quad \zeta \in (0, 1 - \gamma), \quad (9)$$

where  $\bar{A}_j$  is the exogenous fundamental and  $\zeta$  is the agglomeration elasticity. As with the congestion term  $\phi(N_i, L_i)$  in §Municipal Zoning, individual firms are atomistic and take  $A_j$  — hence  $L_j$  — as given when maximizing profit; the externality only appears once the labor-demand condition above is solved out in equilibrium. Substituting (9) into  $L_j = (\gamma A_j / w_j)^{1/(1-\gamma)} \bar{H}_j^F$  and solving explicitly for  $L_j$ ,

$$L_j = \left[ \frac{\gamma \bar{A}_j (\bar{H}_j^F)^{1-\gamma}}{w_j} \right]^{\frac{1}{1-\gamma-\zeta}}. \quad (10)$$

The restriction  $\zeta < 1 - \gamma$  is required for this exponent to be finite and positive, i.e. for equilibrium local labor demand to remain strictly decreasing in  $w_j$  — a stability condition on the strength of the externality relative to the labor share, playing the same role as the  $a > 1$  convexity condition already imposed on  $\phi$ .

## 1.4 Municipal Zoning

Land at each location competes for residential and commercial use. Each municipality  $g \in G$  chooses an iceberg cost  $\{t_i^F\}_{i \in S^g} \geq 0$  on commercial land use at each location: procedural burden — discretionary review, permitting delay, litigation risk, compliance — melts away a fraction of commercial floor space’s usable value before it reaches the market. Because the same underlying land parcel can be put to either use, competitive landowners allocate it to whichever use pays the higher effective return; if both uses are active in equilibrium, this competition equalizes returns at the margin:

$$q_i^R = \frac{q_i^F}{1 + t_i^F}. \quad (11)$$

Combined with the firm’s optimal commercial rent  $q_i^F$  (§Firms) and the residential rent-clearing condition (2), we can get

$$(1 - \gamma) \left( \frac{\gamma}{w_i} \right)^{\gamma/(1-\gamma)} A_i^{1/(1-\gamma)} = (1 + t_i^F) \frac{(1 - \alpha) \sum_j N \pi_{ij} w_j}{\bar{H}_i^R}. \quad (12)$$

The right-hand side is strictly decreasing in  $\bar{H}_i^R$  (holding  $\{\pi_{ij}\}, \{w_j\}$  fixed): raising  $\bar{H}_i^R$  lowers  $q_i^R$  one-for-one with the fixed numerator, by (2). The left-hand side is treated as locally independent of  $\bar{H}_i^R$  at this step by the same simplification already applied to  $w_i$ : since  $A_i = \bar{A}_i L_i^\zeta$  ((9)) and  $L_i$  responds to  $\bar{H}_i^F = \bar{H}_i - \bar{H}_i^R$  through (10), holding  $w_i$  fixed for this step is now understood to hold  $L_i$ , and hence  $A_i$ , fixed as well — a genuine general-equilibrium channel deferred to the quantification stage in the same way as the other GE objects in §Municipal Zoning and §Transportation Authority, not a new approximation. Under this, the implicit function theorem applied to (12) gives

$$\frac{d\bar{H}_i^R}{dt_i^F} = - \frac{q_i^F / (1 + t_i^F)^2}{-\partial[(1 + t_i^F)(1 - \alpha) \sum_j N \pi_{ij} w_j / \bar{H}_i^R] / \partial \bar{H}_i^R} = - \frac{q_i^F / (1 + t_i^F)^2}{-q_i^R / \bar{H}_i^R} = \frac{\bar{H}_i^R q_i^F}{(1 + t_i^F)^2 q_i^R} > 0, \quad (13)$$

using  $q_i^F = (1+t_i^F)q_i^R$  from (11) to simplify. So a higher iceberg cost strictly raises the residential floor-space allocation  $\bar{H}_i^R$  — a marginally higher  $t_i^F$  melts away commercial land's effective return, and competitive land allocation reassigns the freed land to residential use. This confirms  $\bar{H}_i^R(t_i^F)$  is strictly increasing and monotone, so the map between the iceberg-cost instrument and the direct-quantity instrument is one-to-one wherever both uses are active.

The municipality sets  $\{t_i^F\}$  to maximize the total welfare of its residents.<sup>1</sup> In line with the political economy of local land use (Mast, 2024; Rollet and Weiwu, 2025), the municipal government, acting as a representative of its residents, trades off (1) lower rents and reduced worker-congestion from shrinking commercial land — both favor a higher iceberg cost — against (2) the residential crowding that comes with the resulting in-migration, and (3) the loss of local wage income to residents who both live and work at  $i$ , both of which favor a lower iceberg cost.

$$\max_{\{t_i^F\}_{i \in S^g}} W^g = \sum_{i \in S^g} N_i \bar{U}_i \quad \text{s.t.} \quad (11), \quad \bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where  $N_i$  is the equilibrium population at location  $i$  and  $\bar{U}_i = \Gamma(1 - \frac{1}{\epsilon}) \Phi_i^{1/\epsilon}$  is the expected utility of a representative resident at location  $i$ , taking other municipalities' iceberg costs  $\{t_{i'}^F\}_{i' \notin S^g}$  and transportation investment  $\{I_{k\ell}\}$  as given.  $\phi(N_i, L_i)$  is the congestion term that discounts the effective amenity  $B_i(N_i, L_i) = \bar{B}_i/\phi(N_i, L_i)$  introduced in §Households, capturing two sources of local negative externalities: residential congestion from neighbors and the presence of workers near the residential location:

$$\frac{\partial \phi}{\partial N_i} > 0, \quad \frac{\partial \phi}{\partial L_i} > 0,$$

where  $N_i$  denotes residential population and  $L_i$  denotes employment at location  $i$  (defined above). I parametrize the disutility in Cobb-Douglas form:

$$\phi(N_i, L_i) = \eta N_i^a L_i^b, \quad \eta > 0, \quad a, b > 0, \quad (14)$$

so that  $B_i(N_i, L_i) = \bar{B}_i \eta^{-1} N_i^{-a} L_i^{-b}$ : an exogenous amenity level  $\bar{B}_i$ , marked down by local residential and workplace density. Since  $\bar{H}_i^F = \bar{H}_i - \bar{H}_i^R$ , the chain rule reduces the municipal first-order condition to a single comparison,

$$\frac{dW^g}{dt_i^F} = \left( \frac{\partial W^g}{\partial \bar{H}_i^R} - \frac{\partial W^g}{\partial \bar{H}_i^F} \right) \quad (15)$$

$$\times \frac{d\bar{H}_i^R}{dt_i^F} = 0, \quad (16)$$

so the optimum reduces to signing  $\partial W^g/\partial \bar{H}_i^R - \partial W^g/\partial \bar{H}_i^F$ , given  $d\bar{H}_i^R/dt_i^F \neq 0$  ((13) below).

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<sup>1</sup>This differs from Bordeu (2026), who models the local government's objective as maximizing aggregate land value. Under absentee landownership (the assumption maintained throughout this paper, §Environment), land value and incumbent-resident welfare  $\sum_i N_i \bar{U}_i$  are generally distinct objects: the equivalence Bordeu notes holds specifically when residents are also the landowners who capture their own parcel's rent, which is not the case here.

**Channels.** Only the  $i$ -th term of  $W^g = \sum_{i' \in S^g} N_{i'} \bar{U}_{i'}$  depends on  $\bar{H}_i^R, \bar{H}_i^F$  to leading order, holding other locations' land use and the metro-wide index  $\Phi$  fixed. With the congestion term now in the amenity,  $\Phi_i = \bar{B}_i^\epsilon \phi_i^{-\epsilon} \sum_j T_j E_j w_j^\epsilon \tau_{ij}^{-\epsilon} (q_i^R)^{-\epsilon(1-\alpha)}$ , and the partials of  $\bar{U}_i = \Gamma(1 - 1/\epsilon) \Phi_i^{1/\epsilon}$  are exact and closed-form:

$$\frac{\partial \bar{U}_i}{\partial q_i^R} = -\frac{(1-\alpha)\bar{U}_i}{q_i^R}, \quad \frac{\partial \bar{U}_i}{\partial \phi_i} = -\frac{\bar{U}_i}{\phi_i}, \quad \frac{\partial \bar{U}_i}{\partial w_i} = \frac{\pi_{ii|i} \bar{U}_i}{w_i} > 0,$$

where  $\pi_{ii|i}$  is the conditional probability that a resident of  $i$  also works at  $i$  (the local live-work share): the first equality is exact since  $q_i^R$  enters  $\Phi_i$  as a common factor across every  $j$ ; the second is exact because  $\phi_i$  enters  $\Phi_i$  as the single power  $\phi_i^{-\epsilon}$ , common across every  $j$  — no first-order approximation is needed, unlike the earlier income-side formulation where  $\phi_i$  was netted against  $w_j$ ; the third holds because only the  $j = i$  term of  $\Phi_i$  depends on  $w_i$ , so the wage channel is scaled down by how many residents actually capture it, and is likewise simpler than before since  $w_i$  no longer nets against  $\phi_i$ .

Substituting these, together with  $\phi_i = \eta N_i^a L_i^b$ , into  $\partial(N_i \bar{U}_i)/\partial \bar{H}_i^R - \partial(N_i \bar{U}_i)/\partial \bar{H}_i^F$  gives the explicit optimality condition

$$\begin{aligned} & \underbrace{\bar{U}_i \frac{\partial N_i}{\partial \bar{H}_i^R}}_{(1) \text{ in-mover } \geq 0} + \underbrace{N_i \frac{\partial \bar{U}_i}{\partial q_i^R} \frac{\partial q_i^R}{\partial \bar{H}_i^R}}_{(2) \text{ rent } \geq 0} + \underbrace{N_i \frac{\partial \bar{U}_i}{\partial \phi_i} \eta a N_i^{a-1} L_i^b \frac{\partial N_i}{\partial \bar{H}_i^R}}_{(3) \text{ residential congestion } \leq 0} \\ & - \underbrace{N_i \frac{\partial \bar{U}_i}{\partial \phi_i} \eta b N_i^a L_i^{b-1} \frac{\partial L_i}{\partial \bar{H}_i^F}}_{(4) \text{ worker-congestion relief } \geq 0} - \underbrace{N_i \frac{\partial \bar{U}_i}{\partial w_i} \frac{\partial w_i}{\partial \bar{H}_i^F}}_{(5) \text{ local-wage loss } \leq 0} = 0. \end{aligned} \quad (17)$$

Terms (1)–(5) reproduce the trade-off: attracting residents and lowering rents (1)–(2), against the residential crowding those in-movers bring (3), net of the worker-congestion relief from shrinking commercial land (4), shrinking  $\bar{H}_i^F$  contracts local labor demand along an upward-sloping labor supply curve — household workplace choice responds to  $w_i$  with finite elasticity  $\epsilon$  (5). The municipality thus weighs the value of local jobs to its own live-work residents against the congestion those same jobs create. With the productivity agglomeration externality (9), term (5) is larger in magnitude than in a model with exogenous  $A_i$ : shrinking  $\bar{H}_i^F$  lowers  $L_i$  along the labor-demand curve (10) *and* lowers local productivity  $A_i = \bar{A}_i L_i^\zeta$  directly, so  $\partial w_i / \partial \bar{H}_i^F$  now compounds a pure quantity effect with a genuine productivity loss — the local-wage channel through which zoning restriction is costly to residents is strictly stronger once agglomeration is present. Given the monotonicity of  $H_i^R(t_i^F)$ , the municipal first-order condition (15) on  $t_i^F$  is equivalent to the first-order condition on  $\bar{H}_i^R$ .

For now the total land supply is fixed and fully utilized. In a future extension, I will incorporate an endogenous land supply where developers choose whether or not to build.

## 1.5 Transportation Authority

The metropolitan transportation authority chooses investment  $\{I_{k\ell}\}_{(k,\ell) \in E}$  to maximize aggregate household welfare, taking the full zoning vector  $\{\bar{H}_i^R, \bar{H}_i^F\}$  and the route assignment  $\{r_{ij}\}$  as given. Following Bordeu (2026)'s formalization of her local government's problem (her §2.1.4), the maximization is stated with the equilibrium conditions of the rest of the model entered directly as constraints, alongside the budget constraint,

rather than solved for separately beforehand:

$$\max_{\{I_{k\ell}\}_{(k,\ell) \in E}} N \cdot \Gamma\left(1 - \frac{1}{\epsilon}\right) \Phi^{1/\epsilon}, \quad \Phi \equiv \sum_{i'j'} T_{i'} E_{j'} \tilde{V}_{i'j'}^\epsilon \quad (18)$$

subject to (i) household optimality:  $\{\pi_{ij}\}$  given by (3)–(4);

(ii) commuting-cost index:  $\tau_{ij}$  given by (5);

(iii) edge-level commuting cost:  $d_{k\ell} = \exp(\kappa \text{time}_{k\ell}(I_{k\ell}, n_{k\ell}))$  (§Transportation);

(iv) edge traffic:  $n_{k\ell} = \sum_{ij \in S^2} N \pi_{ij} \pi_{ij}^{k\ell}$  (7);

(v) labor market clearing:  $N \sum_i \pi_{ij} = L_j(w_j) \equiv (\gamma A_j / w_j)^{1/(1-\gamma)} \bar{H}_j^F$  (§Firms);

(vi) residential rent clearing:  $q_i^R$  given by (2);

(vii) budget:  $\sum_{(k,\ell) \in E} c_{k\ell} I_{k\ell} \leq K, \quad I_{k\ell} \geq 0,$

where  $c_{k\ell}$  is the per-unit cost of investment on edge  $(k, \ell)$  and  $K$  is the total infrastructure budget. Constraints (ii) and (iii) split what was previously described only in prose as "route choice": (ii) is the bilateral commuting-cost index aggregating route costs within an OD pair (Bordeu 2026's analogous constraint, her eq. 12), and (iii) is the edge-level cost technology itself (her eq. 10) — kept as separate constraints, following her construction exactly, rather than substituted directly, because  $I_{k\ell}$  enters the whole system only through (iii). Constraint (v) folds *firm optimality* directly into *labor-market clearing*: the firm's labor demand  $L_j(w_j)$  is a simple closed-form function of  $w_j$  (§Firms), not an implicit/coupled object, so it is substituted in rather than given its own multiplier — the same combination Bordeu uses for her analogous wage/labor-clearing condition. Constraints (i)–(vi) are solved jointly, not sequentially: (iii) feeds back into (ii) and hence (i), (iv) feeds back into (iii) through  $n_{k\ell}$ , while (v)–(vi) simultaneously pin down  $w_j$  and  $q_i^R$ , which enter  $\tilde{V}_{ij}$  inside  $\Phi$ . This system defines  $\{\pi_{ij}(I), \tau_{ij}(I), d_{k\ell}(I), n_{k\ell}(I), w_j(I), q_i^R(I)\}$  as equilibrium objects rather than free parameters, and hence also  $N_i(I) = N \sum_j \pi_{ij}(I)$  and  $L_i(I) = N \sum_j \pi_{ji}(I)$ ;  $\Phi$  in the objective is the resulting reduced-form value, a function of  $I$  through this whole system, not only through  $\tau_{ij}$  directly.

**Direct edge elasticity.** From  $d_{k\ell} = \exp(\kappa \text{time}_{k\ell})$  and  $\text{time}_{k\ell} = \delta_{k\ell}^0 + \delta_{k\ell}^1 n_{k\ell}^\rho / I_{k\ell}^\beta$  (§Transportation),

$$\frac{\partial \ln d_{k\ell}}{\partial I_{k\ell}} = -\frac{\kappa \beta \delta_{k\ell}^1 n_{k\ell}^\rho}{I_{k\ell}^{\beta+1}} \quad (\text{holding } n_{k\ell} \text{ fixed}), \quad \frac{\partial \ln d_{k\ell}}{\partial n_{k\ell}} = \frac{\kappa \rho \delta_{k\ell}^1 n_{k\ell}^{\rho-1}}{I_{k\ell}^\beta}. \quad (19)$$

**Route-choice bookkeeping.** Differentiating  $\tau_{ij}^{-\theta} = \sum_{r \in R_{ij}} \tau_{ijr}^{-\theta}$  and using  $\ln \tau_{ijr} = \sum_{(k,\ell) \in r} \ln d_{k\ell}$ ,

$$\frac{\partial \ln \tau_{ij}}{\partial d_{k\ell}} = \sum_{r \in R_{ij}} \pi_{r|ij} \frac{\partial \ln \tau_{ijr}}{\partial d_{k\ell}} = \frac{1}{d_{k\ell}} \sum_{r \ni (k,\ell)} \pi_{r|ij} = \frac{\pi_{ij}^{k\ell}}{d_{k\ell}} \quad (\text{holding all other } d\text{'s fixed}),$$

where  $\pi_{r|ij} = (\tau_{ij}/\tau_{ijr})^\theta$  is the conditional route-choice probability and  $\pi_{ij}^{k\ell} = \sum_{r \ni (k,\ell)} \pi_{r|ij}$  is exactly the edge-intensity object of (6): the probability that an  $(i, j)$ -commuter's chosen route uses edge  $(k, \ell)$ . This is the object that links  $\tau_{ij}$  to  $d_{k\ell}$  below; it does not yet involve  $I_{k\ell}$  or  $n_{k\ell}$  directly, since those enter only through  $d_{k\ell}$  itself, constraint (iii).

**Formalizing "subject to equilibrium": the Lagrangian.**  $\tau_{ij}, d_{kl}, w_j, q_i^R, N_i, L_i$  (and, through them,  $\tilde{V}_{ij}$ ) are equilibrium objects, not primitives — they respond to  $I_{kl}$  through the constraint system above. Following Bordeu (2026)'s treatment of her local government's problem (her §2.1.4 and Appendix A.3, which formalizes exactly this "planner subject to equilibrium" structure by entering every equilibrium condition into the Lagrangian with its own multiplier, rather than approximating their effect away), I write the authority's problem as a single Lagrangian over constraints (i)–(vi) of (18) — six multipliers, one per constraint, matching Bordeu's own six ( $\lambda_{ij}, \eta_{Ri}, \eta_{Fj}, \gamma_{ij}, \epsilon_{kl}, \phi_{kl}$  in her notation) rather than substituting any of them away:

$$\begin{aligned} \mathcal{L} = & N\Gamma(1 - \tfrac{1}{\epsilon}) \Phi^{1/\epsilon} - \mu \left( \sum_{kl} c_{kl} I_{kl} - K \right) + \sum_{ij} \lambda_{ij} \left[ T_i E_j \tilde{V}_{ij}^\epsilon - \pi_{ij} \Phi \right] \\ & + \sum_{ij} \gamma_{ij} \left[ \tau_{ij} - ((I - A)^{-1})^{-1/\theta} \right] + \sum_{kl} \psi_{kl} \left[ d_{kl} - \exp(\kappa \text{time}_{kl}(I_{kl}, n_{kl})) \right] \\ & + \sum_{kl} \varphi_{kl} \left[ n_{kl} - \sum_{ij} N \pi_{ij} \pi_{ij}^{kl} \right] + \sum_i \eta_i^q \left[ q_i^R \bar{H}_i^R - (1 - \alpha) \sum_j N \pi_{ij} w_j \right] \\ & + \sum_j \eta_j^w \left[ L_j(w_j) - N \sum_i \pi_{ij} \right], \end{aligned} \quad (20)$$

where  $\lambda_{ij}, \gamma_{ij}, \psi_{kl}, \varphi_{kl}, \eta_i^q, \eta_j^w$  are the multipliers on constraints (i)–(vi) respectively.  $\lambda_{ij}$  is the metro-wide marginal social value of an additional household choosing the  $(i, j)$  commute pattern;  $\gamma_{ij}$  is the shadow value of the commuting-cost index  $\tau_{ij}$ ;  $\psi_{kl}$  is the shadow value of the edge-level cost technology  $d_{kl}$  itself;  $\varphi_{kl}$  is the shadow value of an additional unit of traffic on edge  $(k, \ell)$ .

**First-order conditions for  $I_{kl}$  and  $n_{kl}$ .** With  $d_{kl}$  pulled out as its own variable,  $I_{kl}$  and  $n_{kl}$  each appear in  $\mathcal{L}$  *only* inside their own constraint-(iii) bracket —  $I_{kl}$  nowhere else at all,  $n_{kl}$  also in its own constraint-(iv) bracket. This is strictly simpler than differentiating through every OD pair routed through the edge directly. Differentiating,

$$\mu c_{kl} = -\psi_{kl} \frac{\partial d_{kl}}{\partial I_{kl}} = -\psi_{kl} d_{kl} \frac{\partial \ln d_{kl}}{\partial I_{kl}}, \quad (21)$$

$$\varphi_{kl} = \psi_{kl} \frac{\partial d_{kl}}{\partial n_{kl}} = \psi_{kl} d_{kl} \frac{\partial \ln d_{kl}}{\partial n_{kl}}. \quad (22)$$

Dividing (21) by (22) already eliminates  $\psi_{kl} d_{kl}$  and gives the ratio condition below, *before*  $\psi_{kl}$  itself is pinned down — the level of  $I_{kl}$  relative to  $\varphi_{kl}$  does not require solving the rest of the multiplier system.

**Closing the system: from  $\lambda_{ij}$  to  $\psi_{kl}$ .**  $\psi_{kl}$  is itself determined by  $\mathcal{L}$ 's first-order condition in  $d_{kl}$ , which appears in its own bracket and in every  $\gamma_{ij}$  bracket for OD pairs routed through  $(k, \ell)$  (via  $\partial \ln \tau_{ij} / \partial d_{kl} = \pi_{ij}^{kl} / d_{kl}$ , above);  $\gamma_{ij}$  in turn is determined by the FOC in  $\tau_{ij}$ , which appears in its own bracket and in the  $\lambda_{ij}$  bracket (via  $\tilde{V}_{ij} \propto \tau_{ij}^{-1}$ ). Tracing this chain through — the same linear system flagged in §Transportation Authority's earlier discussion, not expanded symbol-by-symbol here any more than Bordeu expands hers in her main text — reproduces exactly

$$\psi_{kl} d_{kl} = \epsilon \sum_{(i,j):(k\ell) \in r_{ij}} \lambda_{ij} T_i E_j \tilde{V}_{ij}^\epsilon \pi_{ij}^{kl} \equiv \epsilon \Lambda_{kl},$$



so that (21)–(22) read  $\mu c_{k\ell} = -\epsilon(\partial \ln d_{k\ell}/\partial I_{k\ell})\Lambda_{k\ell}$  and  $\varphi_{k\ell} = \epsilon(\partial \ln d_{k\ell}/\partial n_{k\ell})\Lambda_{k\ell}$  — identical in form to a direct-substitution derivation, but now every multiplier in the chain  $(\lambda_{ij} \rightarrow \gamma_{ij} \rightarrow \psi_{k\ell} \rightarrow \varphi_{k\ell})$  is a named, separately-constrained object rather than collapsed into one step. I call  $\Lambda_{k\ell}$  edge  $(k, \ell)$ 's *market access*: the traffic-intensity-weighted sum of accessibility gains it delivers across every OD pair routed through it, each pair weighted by its own shadow value  $\lambda_{ij}$  rather than treated symmetrically. All six multiplier families —  $\lambda_{ij}, \gamma_{ij}, \psi_{k\ell}, \varphi_{k\ell}, \eta_i^q, \eta_j^w$  — solve their own first-order conditions jointly, a linear system coupled because a change in any  $\pi_{ij}$  moves  $N_i, L_i$  (hence the congestion amenity  $B_i$ , hence every  $\tilde{V}_{ij}$ ), every route's cost index, every edge's traffic, every location's rent, and every workplace's wage simultaneously. I do not expand this system symbol-by-symbol here — Bordeu does not either, in her main text — beyond noting it is the same size and type of object as her " $2N + E$ "-plus-route/edge-cost system (her Appendix A.3).

**The optimal-investment condition.** Dividing (21) by (22) eliminates  $\Lambda_{k\ell}$  and gives, using the model's actual  $d_{k\ell}$ ,

$$\mu c_{k\ell} = -\varphi_{k\ell} \frac{\partial d_{k\ell}/\partial I_{k\ell}}{\partial d_{k\ell}/\partial n_{k\ell}} = \varphi_{k\ell} \frac{\beta}{\rho} \frac{n_{k\ell}}{I_{k\ell}}, \quad (23)$$

structurally identical to Bordeu (2026)'s eq. (18). Substituting (22) into (23) and solving for  $I_{k\ell}$ ,

$$I_{k\ell} = \left( \frac{\epsilon \kappa \beta}{\mu} \right)^{1/(\beta+1)} W_{k\ell}^{1/(\beta+1)}, \quad W_{k\ell} \equiv \frac{\delta_{k\ell}^1 n_{k\ell}^\rho \Lambda_{k\ell}}{c_{k\ell}}. \quad (24)$$

Substituting into the budget constraint  $\sum_{k'\ell'} c_{k'\ell'} I_{k'\ell'} = K$  (binding at the optimum, since the marginal benefit in (21) is unbounded as  $I_{k\ell} \rightarrow 0^+$  whenever  $W_{k\ell} > 0$ ) pins down  $(\epsilon \kappa \beta / \mu)^{1/(\beta+1)} = K / \sum_{k'\ell'} c_{k'\ell'} W_{k'\ell'}^{1/(\beta+1)}$ , eliminating  $\mu$  and giving the closed-form optimal investment rule

$$I_{k\ell}^* = K \cdot \frac{W_{k\ell}^{1/(\beta+1)}}{\sum_{(k'\ell') \in E} c_{k'\ell'} W_{k'\ell'}^{1/(\beta+1)}}, \quad W_{k\ell} \equiv \frac{\delta_{k\ell}^1 n_{k\ell}^\rho \Lambda_{k\ell}}{c_{k\ell}}. \quad (25)$$

This is a budget-share rule: edge  $(k, \ell)$ 's share of the infrastructure budget rises with its own  $\lambda$ -weighted market access  $\Lambda_{k\ell}$  and its equilibrium traffic  $n_{k\ell}$  (through  $\rho$ : busier edges get more investment, a "fix the bottleneck" force), and falls with its own cost  $c_{k\ell}$ . It automatically handles the corner case: if  $W_{k\ell} = 0$  (no OD pair routed through  $(k, \ell)$ , or zero equilibrium traffic under  $\rho > 0$ ), then  $I_{k\ell}^* = 0$ , no separate complementary-slackness case needed.

## 1.6 Equilibrium

**Definition 1** (Decentralized Equilibrium). A decentralized equilibrium is a profile

$$\left( \{I_{k\ell}^*\}_{(k,\ell) \in E}, \{t_i^{F*}\}_{i \in S}, \{N\pi_{ij}^*\}_{i,j \in S}, \{n_{k\ell}^*\}_{(k,\ell) \in E}, \{q_i^{R*}, q_i^{F*}, w_i^*\}_{i \in S} \right)$$

such that:

1. **Household optimality.** Commuting flows  $\{N\pi_{ij}^*\}$  are given by (3)–(4) at prices  $(q_i^{R*}, w_i^*)$  and investment  $I^*$ , with route choice by (5) and route-based commuting costs  $\{\tau_{ij}^*\}$ .
2. **Edge traffic.**  $\{n_{k\ell}^*\}$  satisfies (7) given  $\{\pi_{ij}^*\}$  and route assignment  $\{r_{ij}\}$ , jointly with (1) since  $n_{k\ell}^*$  feeds back into  $d_{k\ell}(I_{k\ell}^*, n_{k\ell}^*)$  and hence  $\tau_{ij}^*$ .
3. **Firm optimality.** Wages  $w_i^*$  and commercial rents  $q_i^{F*}$  satisfy the firm's first-order conditions at  $H_i^F = \bar{H}_i^{F*}$ , with labor demand  $L_i^*$  solving (10) under the productivity externality (9).
4. **Market clearing.** Residential floor space clears at  $q_i^{R*}$  given by (2), commercial floor space is fixed at  $H_i^F = \bar{H}_i^{F*}$ , and the labor market clears at  $L_i^* = N \sum_j \pi_{ji}^*$ .
5. **Municipal Nash equilibrium.** Each municipality  $g$  sets the iceberg cost  $\{t_i^{F*}\}_{i \in S^g}$  — equivalently, induces  $\{\bar{H}_i^{R*}, \bar{H}_i^{F*}\}_{i \in S^g}$  via (13) — to satisfy the first-order condition (15), given  $\{\bar{H}_{i'}^{R*}, \bar{H}_{i'}^{F*}\}_{i' \notin S^g}$  and  $I^*$ ; no municipality has a profitable deviation.
6. **Transportation authority optimality.**  $\{I_{k\ell}^*\}$  solves (18), given  $\{\bar{H}_i^{R*}, \bar{H}_i^{F*}\}$ : equivalently,  $I_{k\ell}^*$  satisfies (23) for every edge  $(k, \ell) \in E$ , where the multipliers  $(\lambda_{ij}^*, \gamma_{ij}^*, \psi_{k\ell}^*, \varphi_{k\ell}^*, \eta_i^{q*}, \eta_j^{w*})$  solve their own first-order conditions from (20) jointly with (1)–(4) at the equilibrium profile.

Conditions (v) and (vi) jointly constitute the simultaneous Nash equilibrium between municipalities and the transportation authority: each set of agents takes the other's choices as given and best-responds. The equilibrium iceberg cost  $t_i^{F*} \equiv q_i^{F*}/q_i^{R*} - 1$  is computed from the equilibrium quantities in this profile, not solved for as a separate object.

## 1.7 Social Planner

The decentralized equilibrium splits control of the metro's two instruments across two kinds of agents, each at its own scale: zoning is chosen municipality-by-municipality, each maximizing only over its own  $S^g$ ; investment is chosen by a metropolitan authority, but taking zoning as given. The natural benchmark isolates both distortions at once — a single metropolitan government choosing *both* instruments, for the *whole* metro, jointly. Following Bordeu (2026)'s comparison of her decentralized and "Centralized Planner (Metropolitan)" cases, and using exactly the Lagrangian apparatus of §Transportation Authority, the planner's problem is

$$\begin{aligned}
 & \max_{\{t_i^F\}_{i \in S}, \{I_{k\ell}\}_{(k, \ell) \in E}} N \cdot \Gamma \left(1 - \frac{1}{\epsilon}\right) \Phi^{1/\epsilon} \\
 & \text{subject to} \quad \text{(i)–(vii) of (18), and} \\
 & \quad \text{(viii) land identity: } \bar{H}_i^R + \bar{H}_i^F = \bar{H}_i \quad \forall i \in S.
 \end{aligned} \tag{26}$$

The only difference from (18) is that  $\{\bar{H}_i^R, \bar{H}_i^F\}_{i \in S}$  — equivalently  $\{t_i^F\}_{i \in S}$ , via (11) — are now genuine choice variables for every location, not parameters the constraint system takes as given; correspondingly, (20) gains one term,  $\sum_i \nu_i [\bar{H}_i - \bar{H}_i^R - \bar{H}_i^F]$ , with multiplier  $\nu_i$  on the land identity at  $i$ .

**Investment.**  $I_{k\ell}$  enters the planner’s Lagrangian through exactly the same terms as in (20) — the objective and constraints (i)–(vii) are unchanged — so the FOC is unchanged in form:

$$I_{k\ell}^{SP} = K \cdot \frac{W_{k\ell}^{1/(\beta+1)}}{\sum_{(k'\ell') \in E} c_{k'\ell'} W_{k'\ell'}^{1/(\beta+1)}}, \quad W_{k\ell} \equiv \frac{\delta_{k\ell}^1 n_{k\ell}^\rho \Lambda_{k\ell}}{c_{k\ell}}, \quad (27)$$

identical to (25). The planner’s investment rule is the *same formula*, evaluated at the jointly optimal zoning vector  $\{\bar{H}_i^{R,SP}\}$  instead of the Nash-equilibrium one  $\{\bar{H}_i^{R,DE}\}$  — the entire wedge between  $I_{k\ell}^{SP}$  and  $I_{k\ell}^{DE}$  operates through how zoning shifts  $n_{k\ell}$  and  $\Lambda_{k\ell}$ , not through any change in the investment rule itself.

**Zoning.** The FOC for  $\bar{H}_i^R$  reproduces the same trade-off as (17) — attracting residents against the rent and congestion consequences (channels (1)–(5), unchanged) — but with two additional channels a municipality cannot see, because (17) was derived holding fixed everything outside  $i$ ’s own term of  $W^g = \sum_{i' \in S^g} N_{i'} \bar{U}_{i'}$ , whereas the planner’s objective sums  $\lambda_{i'j'}$  over *every*  $i' \in S$ :

$$\underbrace{\sum_{(i',j'): i' \neq i} \lambda_{i'j'} \frac{\partial [T_{i'} E_{j'} \tilde{V}_{i'j'}^\epsilon]}{\partial \bar{H}_i^R}}_{(6) \text{ cross-jurisdiction spillover}} + \underbrace{\sum_{(k\ell): i \in \{k,\ell\} \text{ or } i \in r_{i',j'} \ni (k\ell)} \frac{\partial \Lambda_{k\ell}}{\partial \bar{H}_i^R} \cdot \frac{\partial I_{k\ell}^{SP}}{\partial \Lambda_{k\ell}}}_{(7) \text{ transit-zoning complementarity}} \quad (28)$$

added to the left-hand side of (17). Term (6) is the welfare gain to residents *outside*  $S^g$  from  $i$ ’s rezoning — lower rents and better access reallocate households metro-wide via the labor and rent markets; under the natural condition that a location’s own rezoning does not make outside residents worse off on net, (6) is weakly positive.

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